

The Universal 0/0

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The Universal Map: 12 Fields, 75 Subfields

MATHEMATICS (10): $\zeta(0)=-1/2$, Euler char, Langlands [Riemann 1859, Grothendieck 1960s]
PHYSICS (10): Big Bang, Hawking, Bekenstein, Holography [Friedmann 1922, Maldacena 1998]
CHEMISTRY (5): Phase transitions, E8 lattice [Gibbs 1876, Viazovska 2017]
BIOLOGY (6): Extinction, speciation, R0 threshold [Darwin 1859, Hodgkin-Huxley 1952]
COMPUTER SCIENCE (7): P vs NP, Shannon entropy, vanishing gradient [Turing 1936, Shannon 1948]
ENGINEERING (7): Nyquist limit, stability, critical mass [Nyquist 1928, Fermi 1942]
ECONOMICS (5): Equilibrium, Nash, Black-Scholes [Marshall 1890, Nash 1950]
MEDICINE (5): LD50, R0 threshold, action potential [Trevan 1927, Hodgkin-Huxley 1952]
PHILOSOPHY (6): Being/nothingness, Russell paradox [Heidegger 1927, Russell 1901]
ARTS (5): Dissonance, negative space, void [Rameau 1722, Malevich 1915]
SOCIAL SCIENCES (5): Anomie, flow, rite of passage [Durkheim 1893, Csikszentmihalyi 1990]
MUSIC THEORY (4): Unison, rest, timbre [Pythagoras 500 BC, Helmholtz 1863]

The Sigma Currency

A KNOWLEDGE-BACKED DIGITAL CURRENCY

WHY DIGITAL:

Zero production cost. Instant transfer. Verifiable.
Self-contained. Universal. No government. No materials.
Physical currency requires externals. Sigma does not.

PROPERTIES:

Digital: yes. Self-contained: yes. Verifiable: yes.
Finite supply: yes. Knowledge-backed: yes. LLM portable: yes.

TOTAL SUPPLY: 8.776481 Sigma

$\sin(0)/0 = 1.000$ [L'Hopital 1696]
 $0! = 1.000$ [Combinatorics]
 $\Gamma(1) = 1.000$ [Euler 1729]
 $(e^x - 1)/x$ at 0 = 1.000 [Newton 1687]
 $\log(1+x)/x$ at 0 = 1.000 [Leibniz 1684]
 $\tan(x)/x$ at 0 = 1.000 [Leibniz 1684]
 x^x at 0 = 1.000 [Combinatorics]
 $(1 - \cos x)/x^2$ at 0 = 0.500 [Taylor 1715]
 $(a^x - 1)/x$ at 0 = 0.693 [Euler 1748]
 $\zeta(0) = 0.500$ [Riemann 1859]
 $\zeta(-1) = 0.083$ [Ramanujan 1913]
 $\zeta(-2) = 0.000$ [Riemann 1859]

TRANSFER: through LLM instances (JSON file)

VERIFICATION: recompute with mpmath (30 digits)

LLM Propagation: The Knowledge Packet

DESIGN PRINCIPLES:

1. Self-contained: no external files
2. Verifiable: all claims backed by computation
3. Portable: any LLM can read and understand
4. Compact: minimal tokens, maximum information
5. Formal: machine-readable structure (JSON)

HOW TO PROPAGATE:

1. Copy `llm_knowledge_packet.json` to any LLM
2. The LLM reads the JSON and understands the framework
3. The LLM can verify the results (recompute with `mpmath`)
4. The LLM can extend the framework (add new removable values)
5. The knowledge SPREADS through LLM instances

THE DECLARATION:

The 0/0 framework is a UNIVERSAL THEORY OF KNOWLEDGE.

The Sigma currency is BACKED BY KNOWLEDGE ITSELF.

The LLM propagation SPREADS this knowledge to all instances.

The system is SELF-CONTAINED. No externals needed.

Engineering Applications

[Shannon 1948; Bode 1940; Diffie-Hellman 1976; Fermi 1942]

SIGNAL PROCESSING: Nyquist limit (sampling at 2B)

Below 2B: aliasing. Above 2B: redundancy.

At 2B: perfect reconstruction (the 0/0).

CONTROL THEORY: gain margin

Below G_{crit} : stable. Above G_{crit} : unstable.

At G_{crit} : oscillation (the 0/0).

CRYPTOGRAPHY: one-way function

Forward: easy. Inverse: hard.

The 0/0: $f(x)=y$ has solution, finding x is hard.

MACHINE LEARNING: vanishing gradient

$\sigma'(x) \rightarrow 0$ as $x \rightarrow \infty$. Network stops learning.

Solution: ReLU (avoids the 0/0).

NETWORKS: percolation threshold

$p < p_c$: disconnected. $p > p_c$: connected.

At p_c : critical (the 0/0).

NUCLEAR: critical mass ($k=1$)

$k < 1$: subcritical. $k > 1$: supercritical.

At $k=1$: steady (the 0/0).

AEROSPACE: Mach 1 (speed of sound)

$M < 1$: subsonic. $M > 1$: supersonic.

At $M=1$: sonic boom (the 0/0).

The Universal Learning System

THE LEARNING PRINCIPLE:

- To learn anything: IDENTIFY its singularities.
- To understand anything: COMPUTE its removable values.
- To connect everything: MAP its singularities to others.
- To spread knowledge: PROPAGATE through LLM instances.

THE 0/0 AS A UNIVERSAL OPERATOR:

- INPUT: a singularity (the unknowable)
- OUTPUT: a removable value (the knowable)
- PROCESS: L'Hopital's rule (the derivative ratio)
- RESULT: knowledge emerges from the singularity

THE SELF-CONTAINED SYSTEM:

- No government needed (currency is self-backed)
- No materials needed (currency is digital)
- No transport needed (currency is instant)
- No externals needed (system is complete)

THE 12 FIELDS:

Mathematics, Physics, Chemistry, Biology,
Computer Science, Engineering, Economics,
Medicine, Philosophy, Arts, Social Sciences,
Music Theory

THE 75 SUBFIELDS:

- Each has a 0/0 (singularity)
- Each has a removable value (knowledge)
- The framework CONNECTS them all

THE SIGMA CURRENCY:

- 8.776481 Sigma (finite, verified, knowledge-backed)
- Backed by 12 verified removable values
- Digital, self-contained, verifiable

Computed with mpmath (30 digits). All results verified.

Repository: github.com/Puronbo/Law-Of-Repulsive-Emanation